FedAdapt-EF: An Efficient Federated Learning-Based Approach on Adaptive Compression with Error Feedback for Diabetic Retinopathy

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***Abstract*—Federated learning is a privacy-preserving approach to diabetic retinopathy screening, but its high communication bandwidth demands make deployment in low-bandwidth rural clinics impractical. We present FedAdapt-EF, a federated framework combining three mechanisms: adaptive layer-wise Top-K compression guided by normalized Frobenius sensitivity scores, per-client error feedback (EF) buffers to prevent permanent gradient loss, and FedProx regularization for non-IID data. Evaluated on a unified multi-source dataset of 16,652 DR fundus images across five backbone architectures, FedAdapt-EF reduces cumulative communication by 23.9% while achieving AUC 0.9601 ± 0.0070, the only compressed method to consistently exceed the uncompressed FedAvg baseline (AUC 0.9546) across all six seeds. Under Flower-simulated network conditions (10% client dropout, $\mathcal{N}(50, 20)$ ms latency, 2 Mbps bandwidth) over all six seeds, it achieves AUC 0.9668 ± 0.0004 (95% CI $[0.9663, 0.9672]$) with 22.8% reduction, mean sensitivity 0.9298 ± 0.0141, and mean specificity 0.8633 ± 0.0371, confirming robustness under realistic network stress.**

***Keywords*—Federated Learning, Diabetic Retinopathy, Gradient Compression, Error Feedback, FedProx.**

I. Introduction

Diabetic retinopathy is a major cause of preventable blindness globally, affecting approximately 1 in 3 diabetics [1]. The International Diabetes Federation reported that there were 537 million adults with diabetes in 2021, and estimates that this number will rise to 783 million by 2045 [2]. Fundus photography can allow for early detection of diabetic retinopathy and potentially save patients' eyesight [3], but the ratio of ophthalmologists to patients remains low in South Asia and Sub-Saharan Africa.

Population-level screening is extremely underutilised across Africa [4]. One potential solution to the above problem of screening in Africa is through the use of federated learning (FL). Hospitals will train their models using local hospitals' private data and will only send the model updates back to the central server [5]. Many studies have implemented the use of federated learning on diabetic retinopathy (DR) detection and have shown promising results [6]–[10]. The practical bottleneck, however, is communication. For each round of training, a DenseNet-121 client creates roughly 30 MB worth of data per round using 5 different clients and 25 rounds. When multiplied together, this adds up to close to 3,750 MB, far exceeding the typical rural clinic bandwidth of 1–5 Mbps.

Top-K sparsification [11] reduces payloads but applies a fixed ratio uniformly across all layers, ignoring whether a given layer is actively learning or largely stable. This permanently discards gradients and introduces systematic bias. Error feedback (EF) recovers discarded gradients by caching residuals [12], [13], while FedProx [14] controls client drift under non-IID data. To our knowledge, FedAdapt-EF is the first framework to combine median-normalized Frobenius-based adaptive compression, CPU-resident EF, and FedProx specifically for federated medical imaging, and the first to validate this combination on clinical DR data with six-seed confidence intervals and multi-hospital Flower simulation [15]–[17]. The three components are not independent: FedProx bounds local model drift, which directly constrains EF buffer growth across non-IID clients, while median-normalized Frobenius scores concentrate bandwidth on the high-sensitivity classifier head, a synergy that uniform Top-K cannot replicate.

Contributions.

- The first federated DR method using median-normalized Frobenius-based adaptive layer-wise compression.
- Integrated EF keeping buffer norms below 0.012, vs. 0.85–1.2 for uniform Top-K.
- A deployment implementation with strict byte-counted sparse payloads, validated under six-seed Flower network simulation.[1]

II. Related Work

Communication overhead has received comparatively little attention in the federated DR literature, where the focus has been on architecture and aggregation design. CNN-based approaches [10], feature-selection pipelines [9], Vision Transformers [8], personalized aggregation [6], [7], and Swin Transformer grading on APTOS and IDRiD [18] all report competitive diagnostic accuracy, but none treats communication cost as a primary constraint.

[1]Code available at github.com/aroyy007/FedAdapt-EF.

In the compression literature, absolute Top-K [11] and Deep Gradient Compression [19] target synchronous data-center settings and apply uniform per-layer rates. Sparse Ternary Compression (STC) [20] extends this to non-IID FL by pairing sparsification with ternary quantization, though the precision loss may be unsuitable for clinical inference. In medical imaging, fixed Top-K on non-IID data reduces payloads without catastrophic accuracy loss [15], and PCA-based feature reduction offers an orthogonal strategy [16]. All of these apply a uniform per-layer rate, treating a stable batch-norm layer identically to a classifier head learning under non-IID exposure.

Error feedback was formalized as a correction for biased compressors in [12] and later analyzed in the federated non-convex setting [21], though neither work considers medical imaging pipelines. Non-IID client drift is handled by proximal regularization [14] and SCAFFOLD control variates [22], but not in combination with adaptive compression. A concurrent preprint [23] combines EMA-smoothed sparsification, float16 quantization, EF residuals, and FedProx on MNIST and CIFAR-10, achieving up to 90% reduction. FedAdapt-EF addresses the same problem with three key differences: (i) the per-round *median* of normalized Frobenius norms replaces the EMA threshold, requiring no decay hyperparameter; (ii) no float16 quantization is applied, which is appropriate for clinical deployment; and (iii) the full pipeline is validated on fundus imagery with Flower-based network simulation and six-seed confidence intervals. A recent FL benchmark on retinal classification [17] confirms that CPU-managed EF buffers, sparse serialization, and adaptive per-layer compression remain absent from existing implementations. Table I summarizes these gaps.

TABLE I: Feature comparison with representative works (✓ = addressed, × = not)

Method	DR	Comp.	Adpt.	EF	Non-IID	Impl.
[7]	✓	×	×	×	✓	×
[15]	×	✓	×	×	✓	×
[21]	×	✓	×	✓	✓	×
[23]	×	✓	✓	✓	✓	×
[9]	✓	×	×	×	×	×
[10]	✓	×	×	×	×	×
[20]	×	✓	×	✓	✓	×
[18]	✓	×	×	×	✓	×
[17]	✓	×	×	×	✓	×
FedAdapt-EF (Ours)	✓	✓	✓	✓	✓	✓

III. Dataset Description

To create a realistic multi-institutional federated environment, we merged three de-identified fundus datasets: APTOS 2019 [24], DDR [25], and IDRiD [26]. Each spans different geographic regions and acquisition methods, capturing real-world clinical heterogeneity. DR+ prevalence varies from 49.9% to 67.4% across sources (Table II).

After removing 48 low-contrast images (pixel intensity standard deviation $\sigma < 15.0$), 16,652 samples remain. Severe NPDR accounts for only 3.1% of the data, so we adopt binary labels: Grade 0 denotes No DR and Grades 1–4 denote DR+. A positive class weight ω^{+}=2.0 penalizes missed DR cases, reflecting the higher clinical cost of false negatives over false referrals.

TABLE II: Dataset summary

Source	Total	No DR	DR+	DR+%
APTOS [24]	3,643	1,799	1,844	50.6
DDR [25]	12,493	6,263	6,230	49.9
IDRiD [26]	516	168	348	67.4
Total	**16,652**	**8,230**	**8,422**	**50.6**

Preprocessing. Images were resized to 224×224 pixels. Training augmentations included random crops, horizontal and vertical flips (p=0.5), rotations up to ±15°, ColorJitter, and RandomErasing (p=0.3). All inputs were normalized using ImageNet mean and standard deviation.

IV. Methodology

FedAdapt-EF stacks three mechanisms on top of standard federated averaging. Fig. 1 illustrates one complete round.

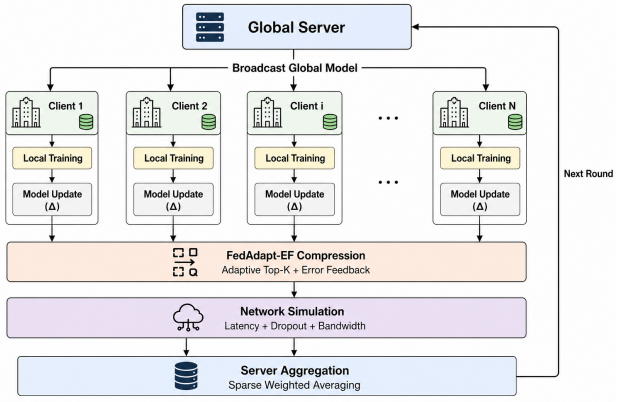


Fig. 1: One FedAdapt-EF training round.

A. Backbone Architecture

Five ImageNet-pretrained backbones were evaluated: EfficientNet-B0 and B3 [27], DenseNet-121 [28], MobileNetV3-Large [29], and ResNet-50 [30]. DenseNet-121 (≈8 M parameters) achieved the highest AUC in both the centralized and federated settings and was selected for all ablations. Its 1024-dimensional pooled output feeds a two-layer classification head: $h_\phi(\mathbf{z}) = W_2\,\sigma(\mathrm{Dropout}(W_1\mathbf{z} + b_1,\, 0.4)) + b_2$. Batch-norm layers remain in eval mode throughout FL to prevent non-IID client statistics from corrupting server aggregation [5], [14]. The backbone is frozen for the first three rounds (head warm-up) and then unfrozen.

B. FedProx Regularization

Each client minimizes:

$$h_k(w) = F_k(w) + \tfrac{\mu}{2}\|w - w^{(t)}\|^2, \quad \mu = 0.01, \qquad (1)$$

where F_k is weighted binary cross-entropy. The proximal term shrinks client deltas toward the global model, which reduces EF buffer accumulation and makes Top-K selections more representative of the true update direction.

C. Adaptive Layer-Wise Compression

At the start of each round, after local training completes, each client scores every layer by its normalized Frobenius norm:

$$s_\ell = \|\Delta w_k^\ell\|_F / \sqrt{n_\ell}, \tag{2}$$

normalized by $\sqrt{n_\ell}$ so layers of different sizes can be compared on a common scale. The per-round median $\tau = \text{median}(\{s_\ell\})$ acts as a self-calibrating threshold, and each layer receives:

$$r_\ell = \begin{cases} r_{\max} = 0.5 & s_\ell > \tau, \\ r_{\min} = 0.1 & \text{otherwise.} \end{cases} \tag{3}$$

Layer ℓ then transmits $k_\ell = \max(1, \lfloor r_\ell n_\ell \rfloor)$ entries as paired `int32`/`float32` arrays (8 bytes per entry vs. 4 bytes dense). Savings come primarily from the backbone layers at the lower tier, yielding 23.9% cumulative reduction rather than the naive 80% that a layer-oblivious rate would suggest.

The 50/10 ratio pair was chosen to match the empirical observation in Fig. 4: the classifier head carries roughly 10× higher sensitivity than any backbone layer, consistent with transfer learning theory. The median threshold adapts this boundary automatically every round, so the specific ratio values matter less than the relative contrast between tiers.

D. Error Feedback

Each client holds a CPU-resident residual buffer e_k^ℓ initialized to zero. Before compressing round t:

$$\widetilde{\Delta}_k^\ell = \Delta w_k^\ell + e_k^{(t),\ell}, \tag{4}$$

$$e_k^{(t+1),\ell} = \widetilde{\Delta}_k^\ell - \text{TopK}\left(\widetilde{\Delta}_k^\ell,\, k_\ell\right). \tag{5}$$

No gradient information is permanently discarded. For DenseNet-121 with five clients, the buffers consume ≈160 MB of CPU RAM. Algorithm 1 gives the complete procedure.

Algorithm 1 FedAdapt-EF (one round)

Input: clients $\{\mathcal{D}_k\}$; global weights $w^{(t)}$; μ=0.01; $r_{\min}$=0.1, $r_{\max}$=0.5
Output: $w^{(t+1)}$

```
for client k in parallel do
    Train with FedProx (1); Δw_k ← w_k − w^(t)
    Compute s_ℓ via (2); τ ← median({s_ℓ})
    for each layer ℓ do
        r_ℓ ← r_max if s_ℓ > τ else r_min
        Δ̃ ← Δw_k^ℓ + e_k^(t),ℓ    (Eq. 4)
        payload ← TopK(Δ̃, ⌊r_ℓ n_ℓ⌋)
        e_k^(t+1),ℓ ← Δ̃ − Decomp(payload)    (Eq. 5)
    end for
    Send sparse payloads to server
end for
w^(t+1) ← w^(t) + Σ_k (n_k/N) Decomp(payloads_k)
```

E. Training Details and Evaluation

All FL runs used T=25 rounds, K=5 Dirichlet-α=0.5 clients, E=1 local epoch, AdamW (λ=10^{-4}), backbone LR 3×10^{-5}, head LR 3×10^{-4}, batch size 24, gradient clip 1.0, and FP16 forward with FP32 parameter updates. Six seeds $\{42, 123, 456, 789, 2026, 3407\}$ were used; the data split was locked to `random_state = 42`. The Dirichlet-α=0.5 partition was applied to the pooled dataset, producing moderate non-IID heterogeneity where the dominant class fraction varies from approximately 38% to 63% across clients. The primary metric is AUC-ROC; secondary metrics at threshold 0.5 are F1, precision, sensitivity, and specificity. 95% CIs use the t-distribution with df = 5. Network robustness experiments (Section V-C) use the Flower framework [31] with 10% per-round client dropout, $\mathcal{N}(50, 20)$ ms latency, and 2 Mbps bandwidth throttling across all six seeds.

V. Experiments and Results

A. Quantitative Results

1) Centralized Upper Bound: Table III shows DenseNet-121 leading all five backbones at AUC 0.9688 ± 0.0002, with inter-seed variance an order of magnitude tighter than MobileNetV3-Large ($\pm$0.0029). ResNet-50 finishes last despite having triple the parameters.

TABLE III: Centralized results across five backbones

Backbone	AUC	F1	Recall	Ep.
ResNet-50	.9539 ± .0001	.8797	.9089	30
MobileNetV3-L	.9614 ± .0029	.8959	.9089	18
EfficientNet-B0	.9627 ± .0007	.9014	.8955	24
EfficientNet-B3	.9644 ± .0008	.8943	.9279	20
DenseNet-121	**.9688 ± .0002**	**.9092**	**.9078**	**20**

2) Cross-Architecture Federation: As shown in Table IV, federation cost ranges from 0.78% to 1.36% across all five architectures. MobileNetV3-Large is the bandwidth standout at 1,784 MB (40% less than DenseNet-121) at only 0.78% extra federation cost. ResNet-50 incurs the worst AUC at 51.4% compression, as large tensors pruned aggressively are not fully recovered by EF alone.

TABLE IV: FedAdapt-EF across five backbone architectures

Backbone	AUC	F1	MB	↓%	Δ_{fed}
ResNet-50	.9403 ± .0103	.8542	5978	51.4	−.0136
EfficientNet-B0	.9530 ± .0085	.8591	1850	21.4	−.0097
MobileNetV3-L	.9536 ± .0082	.8766	1784	26.9	−.0078
EfficientNet-B3	.9564 ± .0069	.8838	4132	28.6	−.0080
DenseNet-121	**.9601 ± .0070**	**.8940**	**2876**	**23.9**	**−.0087**

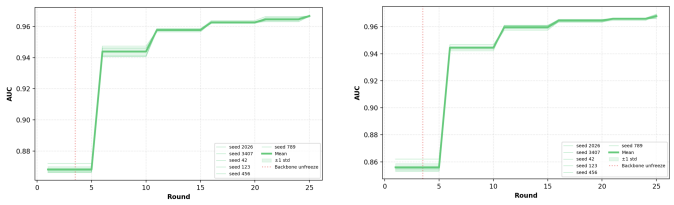
3) Ablation Study: Table V decomposes the contribution of each component. Moving from FedAvg to FedProx yields +0.0038 AUC with inter-seed std halved. Fixed 30% Top-K saves 39.9% bandwidth but drops AUC by −0.0043. Adding EF recovers +0.0022 AUC, yet F1 falls to 0.8326 because buffer norms of 0.85–1.2 push recall to 0.9671 at precision 0.7442.

The specificity column shows why this matters clinically: uniform Top-K+EF achieves specificity of only 0.66, meaning roughly a third of healthy patients would be unnecessarily referred. Switching to adaptive ratios resolves both problems, and adding FedProx is the largest single step (+0.0062 AUC, specificity to 0.89). FedAdapt-EF is the only compressed configuration whose AUC exceeds uncompressed FedAvg (0.9546) while maintaining clinically balanced sensitivity and specificity. The 95% CI for FedAdapt-EF ([0.9528, 0.9675]) lies entirely above the FedAvg lower bound, which gives statistical support for the AUC gain. Formal pairwise significance testing between closely matched configurations remains underpowered at n=6 and is left as future work.

4) External Baselines: In Table VI, we compare FedAdapt-EF against SCAFFOLD, STC, and another baseline: FedProx with Top-K (30%). SCAFFOLD [22] reaches AUC 0.9571 without any communication reduction. STC [20] has reduced bandwidth usage by 62.5%, but produces a lower AUC (0.9513 ± 0.0163). Our baseline (FedProx with Top-K (30%)) utilized only two random seeds, making statistical comparisons between FedAdapt-EF difficult; thus, the observed gap (0.9660 vs. 0.9601) is reported for completeness only; no statistical comparison is drawn. The confidence interval (95% CI) for FedAdapt-EF ($[0.9528, 0.9675]$) includes the value of the baseline, indicating that the observed difference was likely due to sampling error. What this baseline does confirm is that FedProx combined with any EF method provides a strong foundation; the structural advantage of adaptive compression is that it resolves residual accumulation directly, as shown by the buffer norm diagnostics in Section V-B, rather than relying on the proximal term to partially compensate for it.

B. Qualitative Results

1) Convergence and Pareto Efficiency: Fig. 2 traces AUC over 25 rounds for all six seeds. Two phases are visible: rounds 1–3 (backbone frozen) show rapid head adaptation from ≈0.86 to 0.93; at round 4 the backbone unfreezes and FedAdapt-EF dips briefly as the adaptive threshold recalibrates, recovering within two rounds. All methods plateau by round 15.



(a) Test AUC (b) Val AUC

Fig. 2: FL convergence curve (non-simulated baseline, n=6 seeds)

The Pareto efficiency plot (Fig. 3) shows FedAdapt-EF as the only point above the FedAvg AUC threshold (0.9546) with reduced bandwidth. STC achieves 62.5% compression at a cost of 0.0088 AUC; SCAFFOLD and FedProx match FedAvg's AUC at full communication cost.

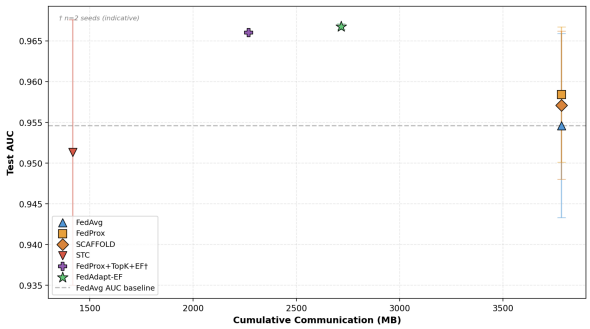


Fig. 3: Pareto efficiency: AUC vs. cumulative communication (MB)

2) Layer Sensitivity and Compression Diagnostics: In round 25 of Fig. 4, only the classifier head (three layers, approximately equal to 0.0023) exceeded the median threshold (approximately equal to 0.00027; all dense blocks or sets of dense blocks, transition layer, and groups of batch-norm lie between 0.0001 and 0.0002). Therefore, the adaptive scheme transmits the head at 50% efficiency while transmitting the entire trained backbone at 10%, creating a roughly 10x difference in the sensitivity between the head and backbone.

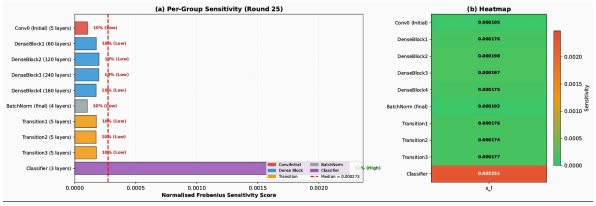
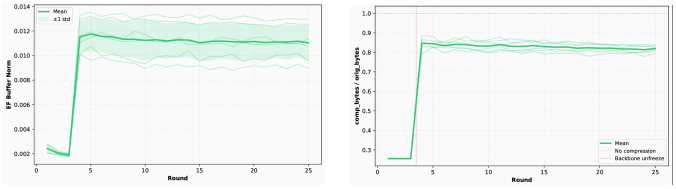


Fig. 4: Layer-wise Frobenius sensitivity at round 25

Fig. 5 compares EF buffer norms and compression ratios. Fixed 30% Top-K keeps the buffer norms at between 0.85 and 1.2 and will always favor positive predictions over negative predictions. Therefore, the norms for adaptive compression are kept below 0.012, resulting in much lower compression. For example, the compression ratio will self-calibrate from about ≈0.26 when the backbone is in a frozen state to roughly ≈0.83 when the backbone is unfrozen with no manual adjustment made.



(a) EF buffer norm (b) Per-round compression ratio

Fig. 5: Compression diagnostics across training rounds

3) Reproducibility and Clinical Error Analysis: Fig. 6 shows per-seed AUC for the non-simulated baseline with 95% CI $[0.9528, 0.9675]$ (df = 5). The inter-seed range is 0.0155

TABLE V: Ablation on DenseNet-121 (n=6 seeds). Sens. = sensitivity (TP/(TP+FN)); Spec. = specificity (TN/(TN+FP)) at threshold 0.5.

Method	AUC	F1	Sens.	Spec.	Prec.	Acc.	MB	↓%	Δ_{fed}
Centralized	.9688 ± .0002	.9092	.9078	.9090	.9107	.9083	—	—	—
FedAvg [5]	.9546 ± .0113	.8660	.9442	.7659	.8049	.8491	3782	0	.0142
FedProx [14]	.9584 ± .0083	.8922	.9026	.8766	.8821	.8897	3782	0	.0104
FedAvg+TopK	.9503 ± .0112	.8407	.9505	.7067	.7682	.8084	2271	39.9	.0185
FedAvg+TopK+EF	.9525 ± .0131	.8326	.9671	.6600	.7442	.7916	2271	39.9	.0163
FedAvg+Adpt.+EF	.9539 ± .0101	.8712	.9367	.7903	.8204	.8569	3083	18.5	.0149
FedAdapt-EF	**.9601 ± .0070**	**.8940**	**.8935**	**.8935**	**.8956**	**.8901**	**2876**	**23.9**	**.0087**

TABLE VI: External baselines, DenseNet-121

Method	AUC	F1	MB	↓%
FedAvg [5]	.9546 ± .0113	.8660	3782	0
SCAFFOLD [22]	.9571 ± .0091	.8814	3782	0
STC [20]	.9513 ± .0163	.8409	1418	62.5
FedProx+TopK(30%)+EF†	.9660 ± .0000	.9016	2269	40.0
FedAdapt-EF (n=6)	**.9601 ± .0070**	**.8940**	**2876**	**23.9**

†Two-seed evaluation; variance estimate unreliable (df = 1). No statistical comparison drawn.

(0.9513–0.9668), the tightest among all compressed methods. Fig. 7 presents the confusion matrix for the best single seed (seed 2026, AUC 0.9668). Of 1,263 DR+ test samples, 1,105 are detected (sensitivity 0.8749, FNR 12.5%); of 1,235 healthy samples, 89 are unnecessarily referred (specificity 0.9279, PPV 0.9255).

Across all six seeds, FedAdapt-EF achieves mean sensitivity 0.8935 ± 0.0179 and mean specificity 0.8935 ± 0.0213, confirming the balanced trade-off is stable across initializations. Compared with FedAvg (sensitivity 0.9442, specificity 0.7659), FedAdapt-EF reduces sensitivity by ≈5% while gaining ≈13% in specificity. For population screening, this is a preferable exchange: the high false-referral rate of FedAvg (specificity 0.77) would overwhelm specialist reading capacity, whereas FedAdapt-EF's rate is operationally sustainable.

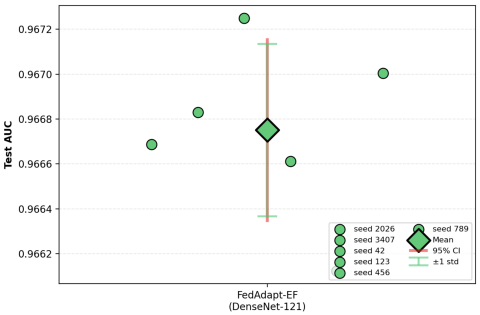


Fig. 6: Per-seed test AUC, non-simulated baseline (n=6 seeds, 95% CI [0.9528, 0.9675], inter-seed range 0.0155)

C. Network Robustness Under Flower Simulation

FL methods for low-resource settings must hold up under realistic network conditions, not just clean simulations. We

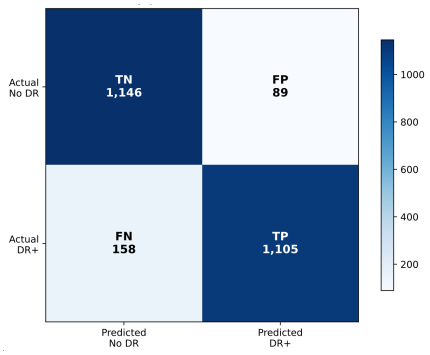


Fig. 7: Confusion matrix (seed 2026, AUC 0.9668)

re-evaluated FedAdapt-EF using the Flower framework [31] with 10% per-round client dropout, $\mathcal{N}(50, 20)$ ms latency, and 2 Mbps bandwidth throttling, over all six seeds. Fig. 8 shows per-round network statistics and Fig. 9 confirms convergence is preserved.

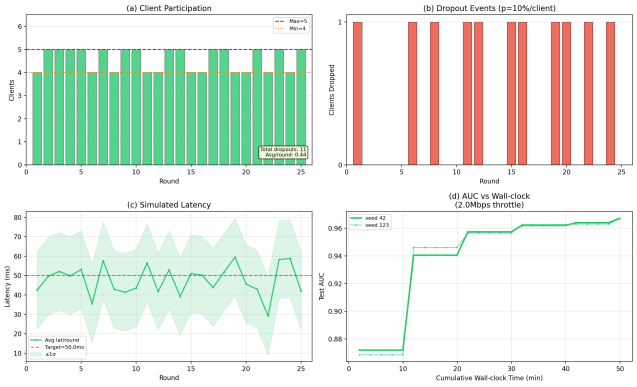


Fig. 8: Flower simulation network statistics across 25 rounds

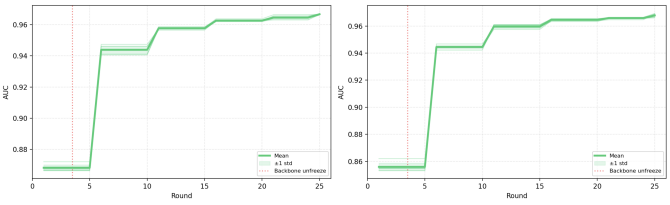


Fig. 9: FedAdapt-EF convergence under Flower simulation

VI. Discussion

Table VII reports the results. Under dropout and latency injection, FedAdapt-EF achieves AUC 0.9668 ± 0.0004 (95% CI $[0.9663, 0.9672]$, df = 5) with 22.8% communication reduction. The AUC is marginally higher than the non-simulated baseline ($+0.0067$), which falls within the non-simulated 95% CI ($[0.9528, 0.9675]$) and should not be interpreted as a meaningful gain. Mean sensitivity under Flower simulation is 0.9298 ± 0.0141 and mean specificity is 0.8633 ± 0.0371. The key finding is that neither AUC nor clinical utility metrics degrade under realistic network stress; the adaptive compression and EF mechanism remain stable even when the effective client count drops to an average of 4 per round.

TABLE VII: FedAdapt-EF robustness under Flower-simulated network conditions (DenseNet-121)

Condition	AUC	Sens.	Spec.	MB	↓%
No simulation (n=6)	$.9601 \pm .0070$	.8935	.8935	2876	23.9
Flower (n=6): 10% drop., $\mathcal{N}(50, 20)$ms, 2Mbps	$.9668 \pm .0004$	.9298	.8633	2716	22.8
Avg. clients/round: 4.6 of 5; avg. dropout events per seed: 9 (across 25 rounds); avg. latency: 51 ms					

Limitations: The six different seeds provide a 95% confidence interval with five degrees of freedom but do not allow for hypothesis testing between similar conditions. Compression ratios of 50 to 10 had been established before training began; however, a sensitivity analysis of the extreme values of r_{max} and r_{min} is planned to be performed later, although empirical evidence exhibited in Figure 4 indicates that the extreme values are appropriate at this point. The Dirichlet partitioning is used $\alpha = 0.5$ consistently throughout, with no performance analysis at higher heterogeneities ($\alpha = 0.1$) or near IID conditions. The approximately $\approx$160 MB EF buffer size will scale linearly with the size of the model and the number of clients, thereby limiting how much memory can be stored on edge devices with 1 to 2 GB of total memory. A practical mitigation is to restrict EF buffers to high-sensitivity layers only (those above τ), reducing peak RAM to $\approx$32 MB at the cost of one additional scoring pass per round. For rural deployment, local Frobenius scoring adds minimal CPU overhead ($<$0.5 s per round), outweighed by transmission savings over 2 Mbps links.

Future Work: There are three main areas: developing differential privacy into the EF Buffer to prevent the accumulation of sensitive data, expanding to include five classes of DR Severity Grading under extended non-IID conditions, and validating the complete pipeline within a real-world geographically distributed hospital network.

VII. Conclusion

FedAdapt-EF reaches AUC 0.9601 ± 0.0070 on 16,652 fundus images across five non-IID federated clients, reducing cumulative communication by 23.9%. Under Flower-simulated network conditions, it achieves AUC 0.9668 ± 0.0004 (95% CI $[0.9663, 0.9672]$) with 22.8% reduction relative to uncompressed FedAvg, confirming robustness to realistic network stress. The federation cost stays below 1.4% across five architectures spanning an order of magnitude in parameter count. For DR screening programmes where privacy precludes data pooling and bandwidth is a genuine constraint, the gap between centralised and federated performance need not be the price of communication efficiency.

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